WORKSHOP 3
COMPUTER VISION

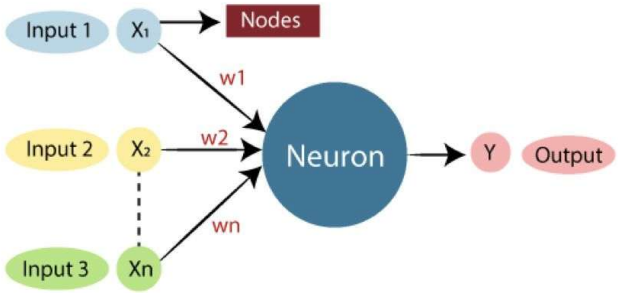


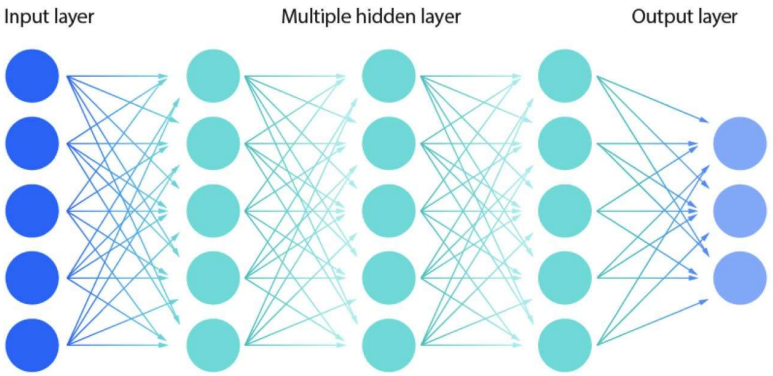
PARAM MANS

RECAP OF WORKSHOP 2

NEURAL NETWORKS:

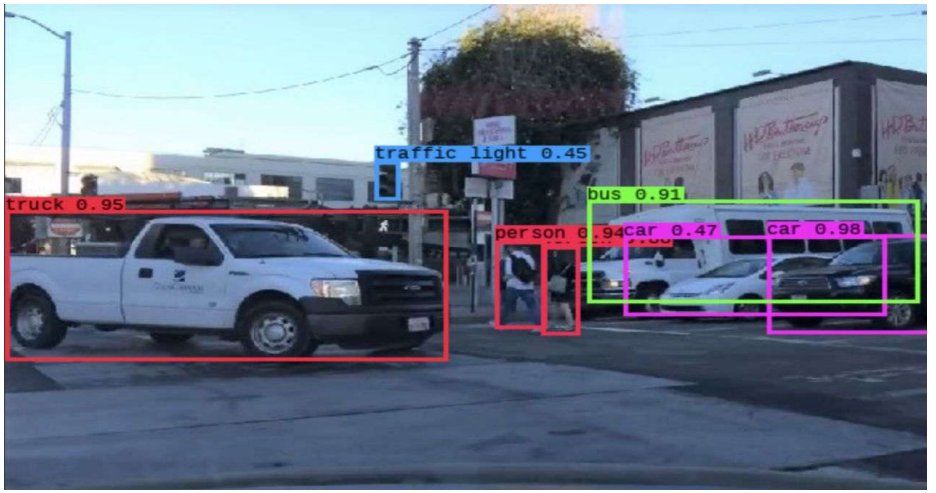
- Modelled on the human brain.
- Sum weighted inputs -> activation function -> output.
- Neurons are organized in **layers.**



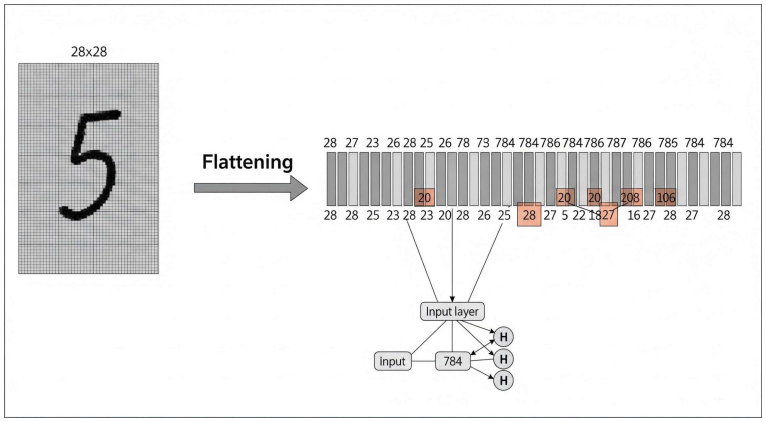


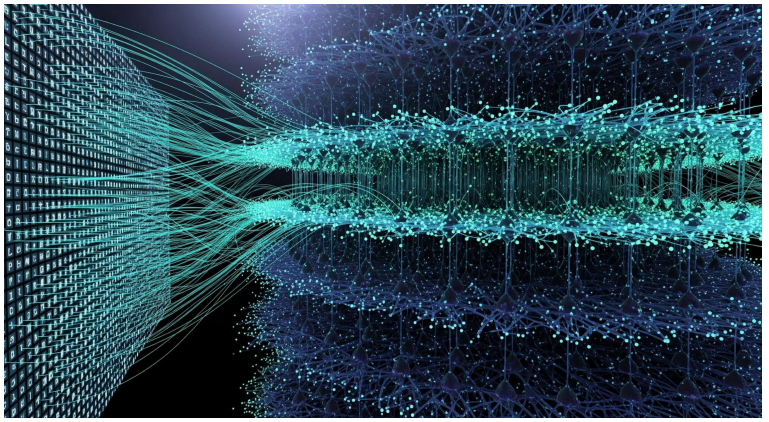
WHAT IS COMPUTER VISION?

- Computers interpreting and processing the world through visual data.
- Object Detection - identifying the location of objects in an image.

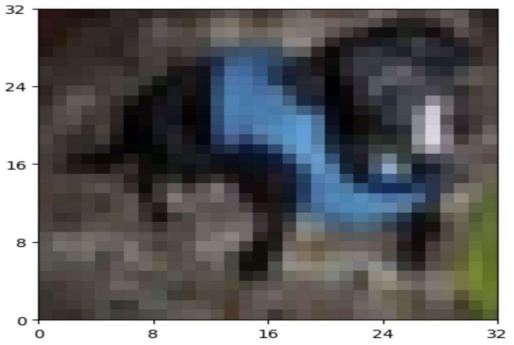


PROBLEMS WITH NEURAL NETWORKS

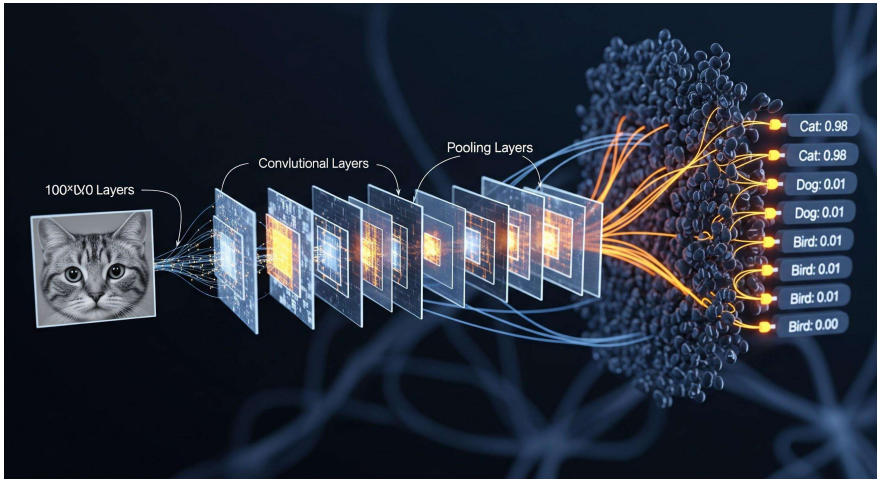




- Unsuitable for images due to spatial information being lost and inefficiency.
- No longer knows that the “eyes” are above the “nose” etc.
- Every pixel connected to every neuron.



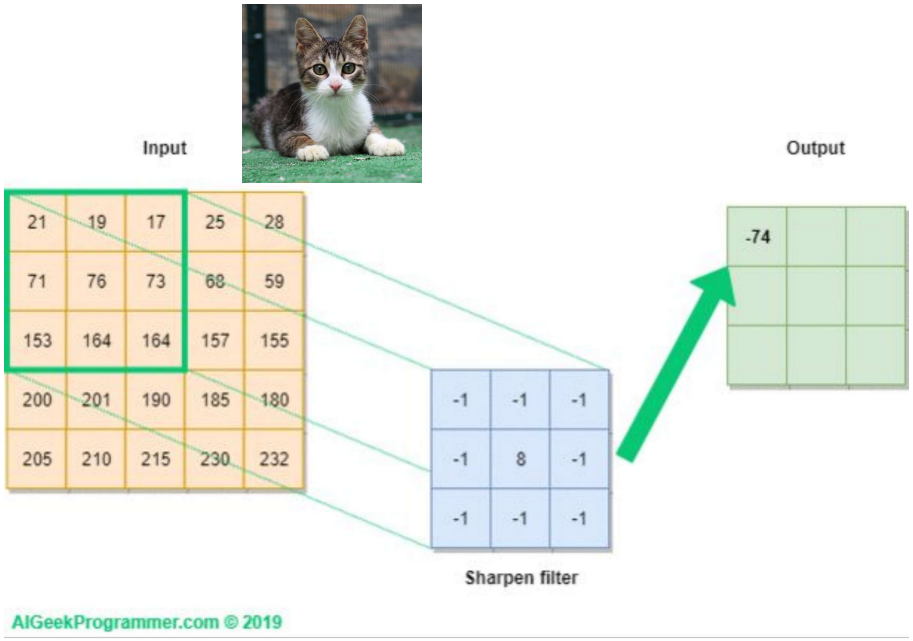
CONVOLUTIONAL NEURAL NETWORKS (CNNs)





- Architecture used for Computer Vision.
- Uses **filters** (like magnifying glasses) to scan image.
- Recognize patterns and clues like edges, textures, and shapes.

CONVOLUTIONAL LAYER

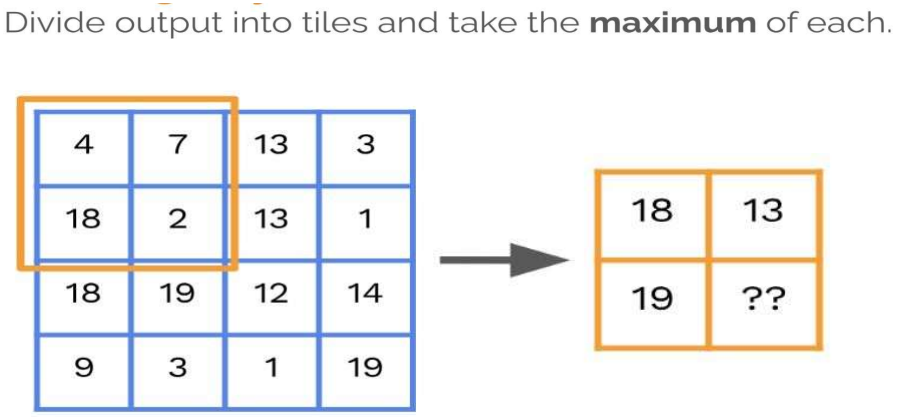




- Layer that lets us find the pattern anywhere at once.
- Different kernels represent different patterns.

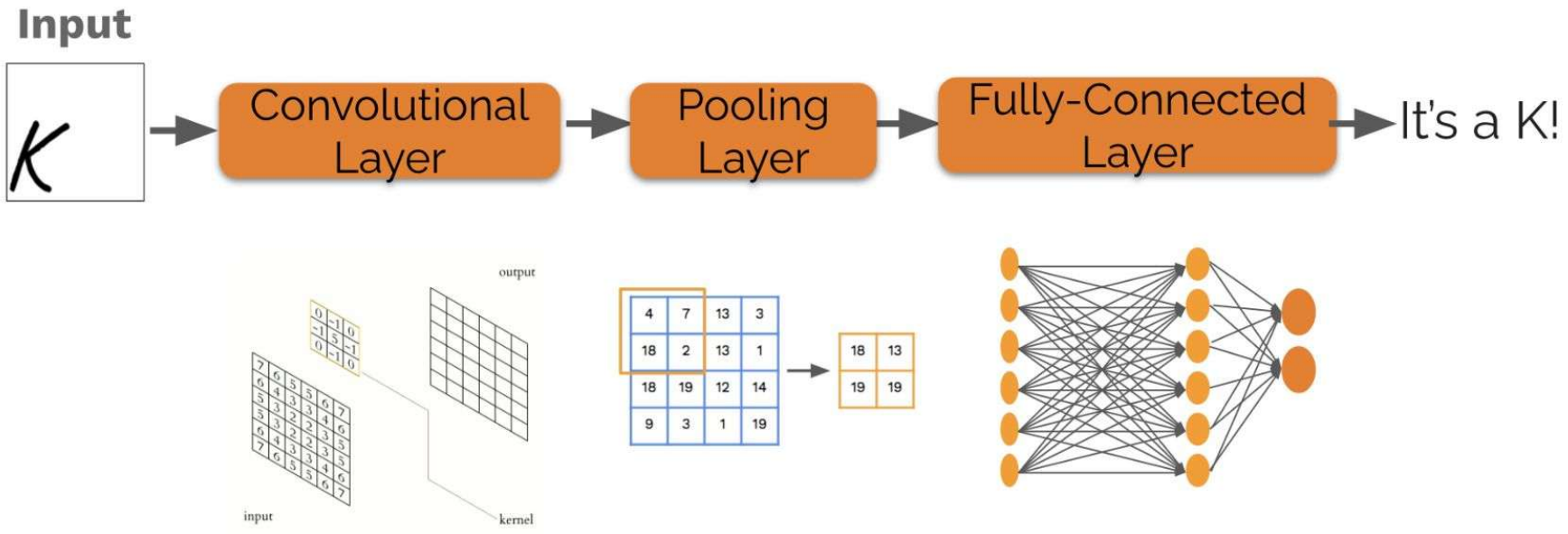
POOLING LAYER

- Used to reduce size and complexity of CNNs.
- Speeds up computation and lowers memory usage.
- Max Pooling (shown) is the most common technique used which selects the maximum value from each region.



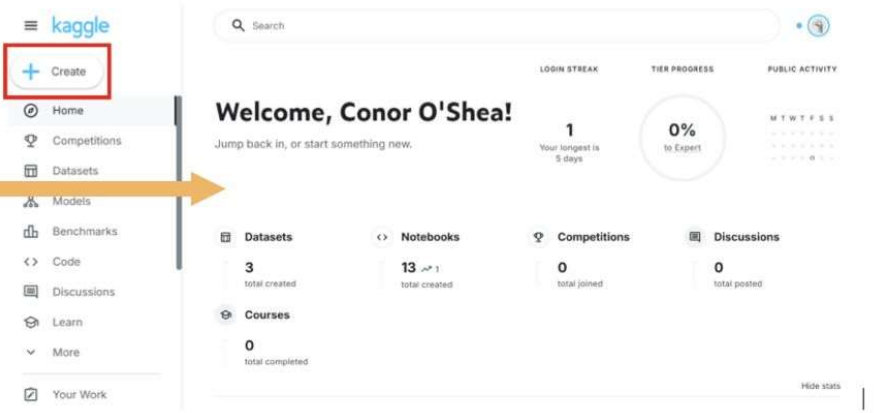
19!

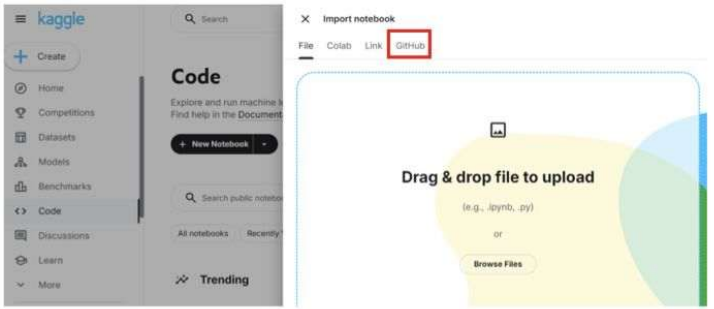
CNN ARCHITECTURE



KAGGLE SETUP FOR CODING

1. Go to: https://www.kaggle.com/
2. Create an account (if you don't have one already)
3. Click the create button in the top left and select "Import Notebook" from the dropdown menu.
4. Select "GitHub"
5. In the search prompt field enter: "EdinburghAI/workshops"
6. In the file prompt select one of the following:





TheoreticalWorkshops/Sem1Workshop2/NeuralNetworks-Beginner.ipynb (26.72 KB)

TheoreticalWorkshops/Sem1Workshop2/NeuralNetworks-Expert.ipynb (25.77 KB)

TheoreticalWorkshops/Sem1Workshop2/NeuralNetworks-Intermediate.ipynb (27.47 KB)

should be Workshop3 instead of Workshop2